Multivar Energy — Use Case: Agentic AI-Driven Grid Resilience & Digitization

Scenario Overview

Multivar Energy is a large green energy grid provider in Europe, delivering sustainable energy to millions of households across 20+ distribution system operators (DSOs) and business units (BUs). With sustainability at the forefront of their mission, Multivar Energy's CIO has secured board-level funding for a comprehensive **data and AI transformation program** — recognizing that the next wave of competitive advantage will not come from data lakes alone, but from **governed, agentic AI systems** that can autonomously reason over unified data, coordinate complex grid operations, and augment scarce engineering expertise.

The CIO's Evolving Mandate

Multivar Energy's CIO is no longer just a technology steward — they sit alongside the CFO and COO as a partner accountable for growth, resilience, and regulatory compliance. The CIO's priorities have shifted from "build a data lake" to "**build an intelligent, agent-ready data platform**" that can:

1. **Power agentic AI workflows** — autonomous agents that can retrieve data, run simulations, coordinate field operations, and recommend actions across the grid — with human operators retaining control over safety-critical decisions
2. **Unify IT and OT data** — bridging the gap between enterprise systems (ERP, CRM, billing) and operational technology (SCADA, EMS, ADMS, AMI, DERMS) into a single governed data estate
3. **Enable compound AI systems** — combining specialized ML models, LLMs, optimization engines, digital twins, and deterministic engineering tools into coordinated workflows rather than relying on a single model
4. **Ensure AI governance and security** — treating AI security as critical-infrastructure security, with full auditability, model monitoring, and human-in-the-loop controls for a regulated utility environment

Why Now: Three Converging Pressures

- **Rapid load growth** from EV adoption, heat pumps, data centers, and industrial electrification is straining grid capacity and making traditional planning cycles too slow
- **Grid modernization urgency** — extreme weather events, distributed energy resources (DERs) like rooftop solar and batteries, and aging infrastructure demand real-time visibility and predictive capabilities
- **AI investment momentum** — 94% of power-and-utility CIOs planned to increase AI investment in 2025, with Gartner forecasting AI adoption in 40% of utility control rooms by 2027

The Challenge: Beyond the Data Lake

Multivar Energy's data lake on **Microsoft Azure** is nascent. While the strategic direction is right, the organization faces the same blockers that prevent most utilities from moving from data collection to data intelligence:

- **Data silos across 20+ DSOs/BUs** — each with different SCADA systems, asset registries, GIS platforms, and reporting formats
- **No unified asset model** — solar panels, substations, transformers, wind turbines, and battery storage systems are tracked in disconnected systems with inconsistent identifiers
- **IT/OT convergence gap** — operational technology data (real-time telemetry, sensor streams, grid events) lives in historians and control systems that don't speak to the enterprise data platform
- **Scalability limitations** — current batch-oriented pipelines cannot support the real-time and near-real-time analytics needed for grid resilience

- **Data governance concerns** — as a regulated utility, Multivar Energy must maintain strict access controls, audit trails, and compliance with EU energy regulations (including NERC-CIP equivalents and GDPR)
- **Fragmented AI efforts** — isolated ML pilots for demand forecasting and asset monitoring exist, but there is no enterprise AI platform, no model governance, and no path from pilot to production

The Vision: An Agentic Grid Intelligence Platform

Rather than simply filling a data lake, Multivar Energy's CIO is championing a **three-horizon transformation**:

Horizon 1: Intelligent Data Foundation (0–9 months)

Build the governed data estate that makes agentic AI possible:

- **Lakehouse architecture on Azure** — Bronze/Silver/Gold medallion layers using Delta Lake, unifying structured data (asset registries, work orders, billing) with semi-structured (sensor telemetry, weather feeds) and unstructured data (inspection reports, drone imagery, maintenance manuals)
- **Unified asset and location model** — a canonical data product that gives every grid asset (transformer, substation, solar panel, battery, feeder) a single identity across all 20+ DSOs
- **Real-time streaming pipelines** — ingesting SCADA, AMI smart meter, IoT sensor, and weather data in near-real-time via Azure Event Hubs and streaming into the lakehouse
- **Unity Catalog for governance** — fine-grained access controls, data lineage, classification (PII, critical infrastructure data), and audit logging across the entire data estate
- **Standardized data products** — governed, documented datasets that DSOs and BUs can consume through self-service analytics, replacing the current patchwork of manual reports

Horizon 2: AI-Powered Decision Support (6–15 months)

Deploy specialized AI models and the first generation of AI assistants:

- **Predictive maintenance models** — combining sensor anomaly detection, failure-history analysis, weather exposure, and asset age to predict transformer, cable, and substation failures before they occur. Priority scoring combines failure probability × consequence × restoration difficulty × customer criticality
- **Demand forecasting and DER optimization** — ML models that forecast load at the feeder level, accounting for weather, EV charging patterns, rooftop solar generation, and battery storage behavior
- **Grid resilience simulation** — "what-if" scenario modeling for storms, equipment failure, cyber incidents, and DER volatility using digital twin representations of the grid
- **AI-powered data agents** — natural-language interfaces that let grid operators, planners, and executives query the lakehouse conversationally: *"Which substations face the highest failure risk during tomorrow's heat event, and what maintenance actions can be completed without reducing reserve margin?"*
- **Computer vision for asset inspection** — analyzing drone and satellite imagery to detect vegetation encroachment, equipment damage, and right-of-way risks

Horizon 3: Agentic AI Operations (12–24 months)

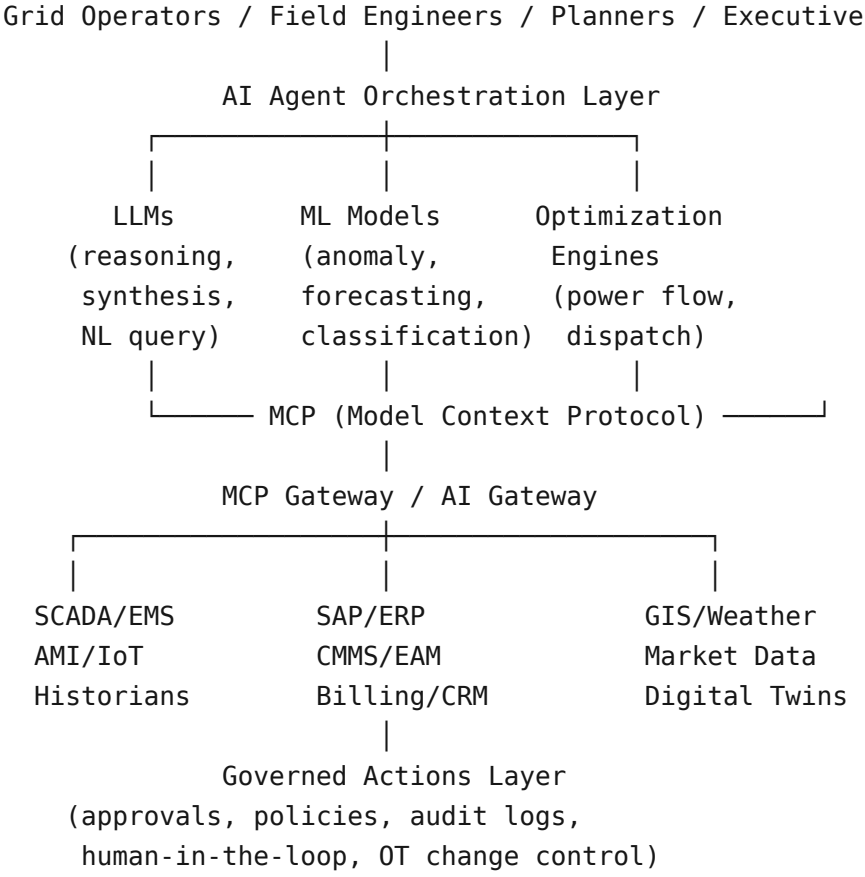
Deploy governed, multi-agent systems that can autonomously coordinate complex grid workflows:

- **Asset Health Agent** — continuously monitors transformer, breaker, cable, and substation telemetry; detects anomalies; estimates remaining useful life; and generates prioritized maintenance recommendations with draft work orders routed to human planners for approval
- **Outage Response Agent** — correlates OMS, AMI, SCADA, weather, vegetation, and crew data in real-time during storm events; proposes switching plans, restoration sequences, and customer communications; tracks restoration dependencies
- **DER Orchestration Agent** — coordinates batteries, EV charging, demand response, solar curtailment, and microgrids within approved operating limits to optimize peak management and renewable integration

- **Grid Planning Agent** — prepares interconnection study cases, runs power-flow and contingency analyses, identifies violations, proposes mitigations, and drafts engineering reports — compressing study timelines from weeks to days
- **Regulatory Compliance Agent** — assembles evidence from operational, environmental, and financial systems; tracks regulatory commitments; and generates auditable filings for EU energy regulators
- **Cyber-Physical Security Agent** — detects unusual behavior across IT and OT environments, enriches alerts with asset context, and recommends containment actions

Agentic AI Architecture: How It Works

The agentic approach uses **compound AI systems** — not a single monolithic model, but an orchestrated ensemble of specialized components:



Model Context Protocol (MCP) serves as the standardized connectivity layer — each enterprise system (historian, SAP, GIS, work-order system, weather service) exposes its data and actions through MCP servers, allowing agents to discover and use them without custom point-to-point integrations. This means:

- Domain teams own their MCP servers and business rules
- AI teams build agents and compound workflows
- Security teams control identity, authorization, and network access
- Operations teams approve production actions

Supervised Autonomy: The Safety Model

For a safety-critical utility, Multivar Energy adopts a **graduated autonomy hierarchy**:

Level	Capability	Example	Human Role

1 — Assist	Summarize, search, explain, draft	"Show me all transformer alerts in Region 3"	User initiates
2 — Recommend	Propose actions with evidence	"Recommend maintenance priority for at-risk assets"	User reviews
3 — Execute with approval	Run simulations, update cases, draft work orders	Agent prepares interconnection study; engineer approves	Explicit sign-off
4 — Bounded automation	Execute pre-approved low-risk actions within strict thresholds	Auto-schedule routine inspections within policy limits	Policy-governed
5 — Autonomous control	Direct grid control	Not targeted in initial deployment	N/A

The practical target for the first 24 months is **Levels 2–4**, with explicit approval gates for any action that could affect grid operations or customer service.

Key Success Metrics

The CIO's dashboard tracks value across four dimensions:

Grid Resilience & Reliability

- SAIDI/SAIFI improvement (System Average Interruption Duration/Frequency)
- Mean time to detect and restore outages
- Interconnection study cycle time reduction
- Renewable curtailment reduction

Operational Efficiency

- 30–40% reduction in unplanned downtime through predictive maintenance
- 15–25% lower maintenance costs through risk-based prioritization
- Engineer/operator hours saved per agent-assisted workflow
- Emergency truck rolls avoided

Data & AI Maturity

- Percentage of grid assets with unified digital identity
- Number of DSOs/BUs consuming standardized data products
- Agent recommendations accepted and correctly executed
- Percentage of AI decisions with traceable evidence

Governance & Compliance

- Human-approval compliance rate for agent actions
- Model drift and exception rate
- Cybersecurity test results for AI systems
- Regulatory filing accuracy and timeliness

Technology Stack on Azure

Capability	Technology
Data storage & lakehouse	Azure Data Lake Storage + Delta Lake (Databricks Lakehouse)
Streaming ingestion	Azure Event Hubs + Structured Streaming
Data governance	Unity Catalog + Microsoft Purview
ML & model operations	Databricks ML Runtime, MLflow, Mosaic AI
Generative AI & agents	Databricks Model Serving, Foundation Model APIs, Agent Framework
Agent connectivity	Model Context Protocol (MCP) servers
AI governance & observability	Unity AI Gateway, MLflow Tracing
Digital twin context	Governed semantic/ontology layer in the lakehouse
Identity & security	Microsoft Entra + NERC-CIP equivalent controls
Visualization	Databricks AI/BI Dashboards, Power BI

Why This Matters for the CIO

This is not a technology project — it is a **business transformation** that positions Multivar Energy to:

1. **Compress decision cycles** — from weeks of manual analysis to minutes of agent-assisted insight
2. **Scale expertise** — augment scarce grid engineers and planners with AI agents that handle routine analysis, freeing humans for judgment-intensive work
3. **Build institutional memory** — capture operational knowledge in governed data products and agent workflows rather than in the heads of retiring engineers
4. **Future-proof the grid** — create an extensible platform that can absorb new DER types, regulatory requirements, and AI capabilities as they emerge
5. **Demonstrate responsible AI leadership** — show regulators, customers, and the board that AI is being deployed with full governance, auditability, and human oversight

The competitive window is closing. The gap between energy companies that commit to agentic AI now and those that hesitate will show up in reliability metrics, operational costs, regulatory standing, and ultimately in the ability to deliver on the green energy mission that defines Multivar Energy.